

MATH 601

Mathematical Foundations of Machine Learning

(2025 Fall Semester)

by
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Prerequisites: I strongly advise you to keep your undergraduate notes on calculus, linear algebra and probability theory nearby.

Credits: (3-0)3 / 7 ECTS

Outline: This will be a *standard* course that reviews mathematical tools necessary to understand the theory behind machine learning. It will have three main parts

- ▶ Linear algebra
- ▶ Multivariate calculus and optimization
- ▶ Probability and information theory

This is the first time I am teaching this course. So the following is merely a tentative outline - some weeks may be light some packed. We'll arrange this weekly plan accordingly.

- Week 1. **Linear Algebra Basics**
- Week 2. **Spectral Theory**
- Week 3. **Singular Value Decomposition**
- Week 4. **Positive Matrices & Perron–Frobenius**
- Week 5. **Calculus Refresher**
- Week 6. **Convex Sets & Functions**
- Week 7. **Convex Optimization**
- Week 8. **Nonconvex Optimization**
- Week 9. **Probability Theory Foundations**
- Week 10. **Concentration Inequalities**
- Week 11. **Advanced Probability for Machine Learning**
- Week 12. **Statistical Estimation**
- Week 13. **High-Dimensional Statistics**
- Week 14. **Information Theory Essentials**

Bibliography:

Core references :

- ▶ Learning Theory from First Principles, Francis Bach. available here, or type :
https://www.di.ens.fr/~fbach/ltfp_book.pdf
- ▶ Deep Learning, Ian Goodfellow, Yoshua Bengio and Aaron Courville. available here or type :
<https://www.deeplearningbook.org/>

Other good references

- ▶ Vershynin, High-Dimensional Probability
- ▶ Boyd & Vandenberghe, Convex Optimization
- ▶ Cover & Thomas, Elements of Information Theory
- ▶ Shalev-Shwartz & Ben-David, Understanding Machine Learning

Programme: Wednesday 12h00 - 15h00 @ H 305

Evaluation:

- Quizzes: % 60
- Final: % 40